

Comparing Native and Non-native English Speakers' Behaviors in Collaborative Writing through Visual Analytics

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Abstract

Understanding collaborative writing dynamics between native speakers (NS) and non-native speakers (NNS) is critical for enhancing collaboration quality and team inclusivity. In this paper, we partnered with communication researchers to develop COALA, an AI-assisted visual analytics tool for comparing NS and NNS behaviors in 162 writing sessions across 27 teams. We validated the effectiveness of COALA through user studies with domain experts (N=2+2) and researchers with relevant experience (N=8). We present the insights discovered by participants using COALA and suggest features for future AI writing assistants for collaborative writing involving NNS.¹

Non-native speakers (NNS) actively engage in collaborative writing: international students write reports with classmates (Cheng, 2013) and advisors (Yang et al., 2021); Wikipedia contributors edit articles in different languages (Hale, 2014); employees in multi-national corporations collaborate on project pages (Atlassian, 2024). Previous research shows that when writing in a non-native language, NNS tend to produce shorter and less complex content compared to writing in their native language (Chenoweth and Hayes, 2001; Wolfersberger, 2003; Uzawa and Cumming, 1989). However, limited research has examined the collaborative writing processes involving NNS and how they write differently from native speakers (NS) (Xiao et al., 2024). Such knowledge will be insightful in enhancing collaboration and fostering inclusive team dynamics involving NNS.

To bridge this gap, in collaboration with communication researchers, we collected document

history and screen recordings of 162 collaborative writing sessions from 27 teams. We are interested in *comparing* authors' behaviors across different linguistic backgrounds and writing stages. For example, what are the common behavior patterns of NS and NNS, respectively? How do they differ in the early and late stages of collaborative writing?

However, existing document visualization and analytics tools fall short of supporting our analysis goal. Visualizations of document versions have long been used to investigate the dynamics of collaborative writing. For example, History Flow (Viégas et al., 2004) uses a Sankey diagram to encode content contributions from different Wikipedia authors across different versions; Time Curves (Bach et al., 2015) project different document versions on a 2D curve based on their similarity and temporal order. However, such visualizations focus on the document content instead of the writers' behaviors during the writing process, such as browsing the internet or using translation tools.

Event sequence visualization tools are better suited for behavioral data but still do not effectively meet our needs. For example, TipoVis (Han et al., 2015) allows comparisons between two sequences at a time, while our analysis requires comparing behavior sequences across multiple authors and sessions. CoCo (Malik et al., 2015) supports comparing event sequences belonging to different cohorts, but focuses primarily on aggregated metrics such as frequencies. In our case, we need to identify behavioral differences between NS and NNS at multiple levels of granularity with meaningful qualitative descriptions.

Furthermore, to effectively analyze such complex behavioral datasets, it is important to combine visualizations with automated methods such as data mining and clustering. Previous research highlights challenges related to the interpretability and trustworthiness of automated meth-

¹This extended abstract summarizes our work recently accepted by CHI 2025

ods (Chuang et al., 2012; Chatzimparmpas et al., 2020; Liang et al., 2023; Yang et al., 2020). As communication research experts, our collaborators possess contextual knowledge about their dataset and multilingual communication in general but are not familiar with the mechanisms of automated methods. Before incorporating the model-generated results into their analysis, they must interpret and trust them.

In comparing the behaviors of NS and NNS in collaborative writing, we thus face two challenges: 1) the limitations of existing text and event sequence visual analytics approaches, and 2) the lack of interpretability and trust in automatic data analytics. To address these challenges, we worked closely with the communication experts to formulate data models and task requirements, iterated on visualization designs to assess their applicability and scalability, and identified factors that might hinder interpretation and trust. We made the following contributions:

- COALA, a visual analytics tool for comparing native and non-native speakers' behaviors in collaborative writing. COALA displays the uncertainties of multiple clustering results and supports interactive refinement of clusters while leveraging large language models (LLM) to generate cluster summaries.
- Empirical validation of COALA through a focus group session (N=2+2) and individual study sessions with researchers in related fields (N=8), where the participants used COALA to analyze behavioral differences between NS and NNS.
- Findings that inform future AI writing assistants for collaborative writing involving NNS, including detecting diverse contribution types, reducing context switching, providing guardrails for machine translation, and summarizing collaborators' activities.

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